

RACTF 2020 Crypto - 04

deuterium — 2020-06-09

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/ractf/crypto/04/2020-06-09-RACTF-2020-04.html>

Abstract

CTF writeup: RACTF 2020's 04 shuffles columns but leaves English letter frequencies on the nametag. A visual solver undoes the columnar transposition.

04

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IOINR EANED GTTEA INEHM ETEIS
PMAON RGNHB BTTWE EEYSC YMOET
PTOAL STEII DAHDE LTNUT MYSOR
OTIEH WEGTN TMHAO YSYWW DNOAA
EDHNG LNNHI OHDGE SSEOT YAEAD
TENRY TITBO TRDHI OMALN DBTOO
DEAUH EFTOR LREVO EHSMS CWEST
AAADY EAIZN RRULT JNTWY SLA00
ATTHK UTYED GOOMY F0SEF SASD0
NSNAH OSSIO EIBPS ALCCR NDTNO
EMEHA STNEL TIHIE AABED PYTMP
SO0ED IUWHS LACEE TSORS NUICY
RASMU TEEMR NSTME EYUXG STA0U
0SE0Y BIELR YNLEW CUTID THNES
PGMKE YOYNE YEVYE UTSTY NOSEY
RIAER NDSEC OLCRX XOYCE X
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The flag is the 2nd and 3rd 9 letter words concatenated together

The first thing while dealing with simple ciphers is checking letter frequency. Heading to frequency analysis section of dcode.fr —Letter—Count—Percentage— —:—:—:—:—:—

—E— 53×—13.38%— —T— 38×—9.6%— —O— 34×—8.59%— —S— 31×—7.83%—
—A— 28×—7.07%— —N— 27×—6.82%— —Y— 23×—5.81%— —I— 20×—5.05%—
—D— 18×—4.55%— —H— 17×—4.29%— —R— 17×—4.29%— —M— 14×—3.54%—
—L— 13×—3.28%— —U— 11×—2.78%— —C— 10×—2.53%— —G— 8× —2.02%—
—W— 8× —2.02%— —B— 7× —1.77%— —P— 6× —1.52%— —X— 4× —1.01%—
—F— 3x —0.76%— —V— 3× —0.51%— —K— 2× —0.51%— —Z— 1x —0.25%— —J—
1x —0.25%—

This cipher turned out to be Columnar Transposition, A good visual solver for the same turns out to be [this site](#)

Proposed Key length: (re)load table

Now try to arrange these to form words (by clicking and dragging the table numbers). The text box below shows the output if you tried to decrypt with this key. If you think the period is wrong simply change the number, and press reload.

9	1	7	3	0	8	2	6	10	4	5
D	O	N	N	I	E	I	A	G	R	E
E	T	H	A	T	M	E	E	T	I	N
G	I	N	P	E	R	S	O	N	M	A
Y	B	E	T	H	E	B	E	S	T	W
A	Y	T	O	C	O	M	P	L	E	T
E	T	H	I	S	D	E	A	L	I	D
O	N	O	T	T	R	U	S	T	M	Y
M	E	N	W	I	T	H	T	H	E	G
O	O	D	S	A	N	Y	W	A	Y	W
H	E	N	H	A	N	D	L	I	N	G
T	H	E	G	O	O	D	S	Y	E	S
T	E	R	D	A	Y	A	N	I	T	E
M	B	I	T	T	O	O	H	A	R	D
A	N	D	B	L	E	D	O	U	T	O
V	E	R	T	H	E	F	L	O	O	R
T	H	E	M	E	S	S	W	A	S	C
R	A	Z	Y	A	N	D	I	R	E	A
L	L	Y	J	U	S	T	W	A	N	T
Y	O	U	T	O	T	A	K	E	T	H
E	G	O	O	D	S	O	F	F	M	Y
H	A	N	D	S	A	S	S	O	O	N
A	S	P	O	S	S	I	B	L	E	I
M	C	O	N	C	E	R	N	E	D	T
H	A	T	T	H	I	S	L	I	N	E
M	A	Y	B	E	T	A	P	P	E	D
S	O	W	E	S	H	O	U	L	D	I
N	C	R	E	A	S	E	O	U	T	S
E	C	U	R	I	T	Y	M	E	A	S
U	R	E	S	M	Y	N	E	X	T	M
E	S	S	A	G	E	T	O	Y	O	U
W	I	L	L	B	E	E	N	C	R	Y
P	T	E	D	U	S	I	N	G	T	H
E	K	E	Y	M	Y	E	N	V	O	Y
S	E	N	T	Y	O	U	Y	E	S	T
E	R	D	A	Y	S	I	N	C	E	R
E	L	Y	R	O	C	C	O	X	X	X

DONNIEIAGREETHATMEETINGINPERSONMAYBETHEBESTWAYTOCOMPLETETHISDEALIDONOTTRUSTMYMENWITHTHEGOODSANYWAYWHENHANDLINGTHEGOODSYESTERDAYANITEMBITTOOHARDANDBLEDOUTOVERTHEFLOORTHEMESSWASCRAZYANDIREALLYJUSTWANTYOUTOTAKETHEGOODSOFFMYHANDSASSOONASPOSSIBLEIMCONCERNEDTHATTHISLINEMAYBETAPPEDSOWESHOULDINCREASEOUTSECURITYMEASURESMYNEXTMESSAGE TOYOU WILL BE ENCRYPTED USING THE KEY MY ENVOYS ENT YOU YESTER DAYS SINCE

DONNIEIAGREETHATMEETINGINPERSONMAYBETHEBESTWAYTOCOMPLETETHISDEALIDONOTTRUSTMYMENWITHTHEGOODSANYWAYWHENHANDLINGTHEGOODSYESTERDAYANITEMBITTOOHARDANDBLEDOUTOVERTHEFLOORTHEMESSWASCRAZYANDIREALLYJUSTWANTYOUTOTAKETHEGOODSOFFMYHANDSASSOONASPOSSIBLEIMCONCERNEDTHATTHISLINEMAYBETAPPEDSOWESHOULDINCREASEOUTSECURITYMEASURESMYNEXTMESSAGE TOYOU WILL BE ENCRYPTED USING THE KEY MY ENVOYS ENT YOU YESTER DAYS SINCE RLYROCCOXX

Fixing spaces in the plaintext, we get,

DONNIE I AGREE THAT MEETING IN PERSON MAY BE THE BEST WAY TO COMPLETE
THIS DEAL I DO NOT TRUST MY MEN WITH THE GOODS ANYWAY WHEN HANDLING THE
GOODS YESTERDAY AN ITEM BIT TOO HARD AND BLED OUT OVER THE FLOOR THE MESS
WAS CRAZY AND I REALLY JUST WANT YOU TO TAKE THE GOODS OFF MY HANDS AS
SOON AS POSSIBLE I M CONCERNED THAT THIS LINE MAY BE TAPPED SO WE SHOULD
INCREASE OUT SECURITY MEASURES MY NEXT MESSAGE TO YOU WILL BE ENCRYPTED
USING THE KEY MY ENVOY SENT YOU YESTERDAY SINCERELY ROCCO XXX

This is the list of all 9 letter words, we have to take 2nd and 3rd to form the flag

['YESTERDAY', 'CONCERNED', 'ENCRYPTED', 'YESTERDAY', 'SINCERELY']

CONCERNEDENCRYPTED

Here is a better analysis of ciphers 01-06 [Youtube Link](#)

| Interactive controls are omitted from this PDF. See the canonical HTML version.